HKUST MATH 2431 - problem set 10

1.

1. (∗) **Multivariate normal distribution** [4 Points]

Let $X = (X_1, ..., X_d)^\top$ be a random vector with values in $(\mathbb{R}^d, \mathscr{B}(\mathbb{R}^d))$. We say that X has the d-dimensional multivariate normal distribution with $\mu \in \mathbb{R}^d$ and $\Sigma \in \mathbb{R}^{d\times d}$, where Σ is symmetric ($\Sigma^\top = \Sigma$) and positive definite ($u^\top \Sigma u \geq 0$ for all $u \in \mathbb{R}^d$ and $u^\top \Sigma u \Leftrightarrow u = 0$) if it is continuously distributed with the probability density function

$$f(x) = \frac{1}{(2\pi)^{\frac{d}{2}}\sqrt{\det \Sigma}} \exp\left(-\frac{1}{2}(x-\mu)^\top \Sigma^{-1}(x-\mu)\right), \qquad x = (x_1, ..., x_d)^\top \in \mathbb{R}^d.$$

We write $X \sim \mathcal{N}_d(\mu, \Sigma)$.

(a) Show that f is indeed a probability density function.

Hint: Use from linear algebra that a positive definite symmetric matrix Σ is diagonalizable and has strictly positive eigenvalues. You can also use without proof the following special case of the multidimensional transformation rule:

$$\int_{\mathbb{R}} g(x)\mathrm{d}x = \int_{\mathbb{R}^d} g(By)\det(B)\mathrm{d}y$$

for an invertible positive definite matrix B, if the integrals exist.

1.a. check non-negativity: easily follows from that $\exp > 0$,

$\underbrace{\det \Sigma}_{= \text{product of eigenvalues}} > 0$ as Σ is $\underbrace{\text{positive definite}}_{\text{all eigenvalues are positive}}$.

check unity:

$$\frac{1}{(2\pi)^{\frac{d}{2}}\sqrt{\det\Sigma}} \underbrace{\int_{\mathbb{R}^d} \exp\left(-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\right)}_{\text{need to show } = (2\pi)^{\frac{d}{2}}\sqrt{\det\Sigma}} dx^n \stackrel{?}{=} 1$$

apply spectral theorem for real sym matrix: $\Sigma = P^T \underset{\substack{\text{diagonal matrix} \\ \text{of eigenvalues} \\ > 0}}{D} P$ (note, $P^{-1} = P^T$, and $\det P = \det P^{-1} = \pm 1$)

$$\Sigma^{-1} = P^T D^{-1} P$$

$$(x-\mu)^T \Sigma^{-1} (x-\mu)$$
$$= (P(x-\mu))^T D^{-1} (P(x-\mu))$$

$$\int_{\mathbb{R}^d} \exp\left(-\frac{1}{2}(P(x-\mu))^T D^{-1} (P(x-\mu))\right) dx^d$$

$$= \int_{\mathbb{R}^d} \exp\left(-\frac{1}{2} y D^{-1} y\right) dy^d \quad y = g(x) = P(x-\mu)$$

and $|\det g(x)| = 1$ because $|\det P| = 1$

$$= \int_{\mathbb{R}^d} \exp\left(-\frac{1}{2}\sum_{k=1}^d \frac{y^2}{\lambda_k}\right) dy$$

since D^{-1} is diagonal)

$$= \prod_{k=1}^d \underbrace{\int_{\mathbb{R}} \exp\left(-\frac{1}{2}\frac{y^2}{\lambda_k}\right) dy}_{= \sqrt{2\pi\lambda_k} \text{ by PDF of } N(0, \lambda_k)}$$

$$= \sqrt{(2\pi)^d \prod_{k=1}^{d} \lambda_k} = (2\pi)^{\frac{d}{2}} \sqrt{\underbrace{\det \Sigma}_{=\text{product of eigenvalues}}}$$

(b) Show that $X_1, ..., X_d$ are independent and normally distributed with $X_i \sim \mathcal{N}(\mu_i, \sigma_i^2)$ if and only if $X \sim \mathcal{N}_d(\mu, \Sigma)$, where $\Sigma = \mathrm{diag}(\sigma_1^2, ..., \sigma_d^2)$.

1.b. both directions easily seen from the equality

$$\left.\prod_{i=1}^{d} \frac{1}{\sqrt{2\pi\sigma_k^2}} \int_{\mathbb{R}} \exp\left(-\frac{1}{2} \frac{(x-\mu_k)^2}{\sigma_k^2}\right) dx \right\}$$

interpreted as joint PDF of mutually independent $X_i \sim N(\mu_i, \sigma_i^2)$

$$= \frac{1}{(2\pi)^{\frac{d}{2}} \sqrt{\prod_{k=1}^{d} \sigma_k^2}} \int_{\mathbb{R}^d} \exp\left(-\frac{1}{2} \sum_{k=1}^{d} \frac{(x_k - \mu_k)^2}{\sigma_k^2}\right) dx$$

$$\left.= \frac{1}{(2\pi)^{\frac{d}{2}} \sqrt{\det \Sigma}} \int_{\mathbb{R}^d} \exp\left(-\frac{1}{2} (x-\mu)^T \Sigma^{-1} (x-\mu)\right) dx \right\}$$

interpreted as $X \sim N_d(\mu, \Sigma)$ where Σ is a diag matrix

where $\Sigma = \mathrm{diag}(\sigma_1^2, ..., \sigma_d^2)$ and $\mu = (\mu_1, ..., \mu_d)$

prop. of diagonal matrix used:

$\det \Sigma$ = product of diagonal elements

$$\Sigma^{-1} = \mathrm{diag}\left(\frac{1}{\sigma_1^2}, \dots \frac{1}{\sigma_d^2}\right)$$

$$(x-\mu)^{-1} \Sigma^{-1} (x-\mu) = \sum_{k=1}^{d} \frac{(x_k - \mu_k)^2}{\sigma_k^2}$$

(c) Show that for $i, j = 1, ..., d$, one has $\mathbf{E}[X_i] = \mu_i$ and $\mathrm{Cov}[X_i, X_j] = \Sigma_{i,j}$, using the rule $(\star)$ below.

1.c. For $E[X_i] = \mu_i$, Take $A = (0, \dots, 0, 1, 0, \dots, 0)$ as a row vector
simply $\quad$ i-th element.

For $\mathrm{Cov}[X_i, X_j] = \Sigma_{i,j}$.
For $i \neq j$,
Use $\mathrm{Var}(X_i + X_j) = \mathrm{Var}(X_i) + \mathrm{Var}(X_j) + 2\,\mathrm{Cov}[X_i, X_j]$

$\Rightarrow \mathrm{Cov}[X_i, X_j] = \frac{1}{2}\left(\mathrm{Var}(X_i + X_j) - \mathrm{Var}(X_i) - \mathrm{Var}(X_j)\right)$

$\mathrm{Var}(X_i + X_j) = \Sigma_{i,i} + \Sigma_{i,j} + \Sigma_{j,i} + \Sigma_{j,j}$
by taking $A = (0, \dots, 0, 1, 0, \dots, \dots, 0, 1, 0, \dots, 0)$
i-th pos $\quad$ j-th pos

$\mathrm{Var}(X_i) = \Sigma_{i,i}$ $\quad$ by taking $A = (0, \dots, 0, 1, 0, \dots, 0)$
$\mathrm{Var}(X_j) = \Sigma_{j,j}$ $\quad$ i/j-th pos

$\mathrm{Cov}[X_i, X_j] = \frac{1}{2}\left(\Sigma_{i,j} + \Sigma_{j,i}\right)$
$= \Sigma_{i,j}$ $\quad$ ($\Sigma_{i,j} = \Sigma_{j,i}$ by sym of Σ)

For $i = j$, taking $A = (0, \dots, 0, 1, 0, \dots, 0)$
i-th pos
yields the result $\mathrm{Cov}[X_i, X_i] = \mathrm{Var}(X_i) = \Sigma_{i,i}$ directly.

(d) Prove ($\star$) for $d = 2$.

Rule ($\star$): Let $X \sim \mathcal{N}_d(\mu, \Sigma)$ with $\mu \in \mathbb{R}^d$ and $\Sigma \in \mathbb{R}^{d\times d}$ symmetric and positive definite. Let $p \leq d$, $A \in \mathbb{R}^{p\times d}$ with full rank p, and $b \in \mathbb{R}^p$. Then $AX + b \sim \mathcal{N}_p(A\mu + b, A\Sigma A^\top)$.

1.d. for $d=2$, $p=1$ or 2

when $p=1$ $\quad Ax+b = A_1x_1 + A_2x_2 + b$

$\mu' = E[Ax+b] = A_1\mu_1 + A_2\mu_2 + b$

$= A\mu + b$

$\Sigma' = Var(Ax+b) = A_1^2 Var(x_1) + 2A_1A_2 Cov(x_1,x_2) + A_2^2 Var(x_2)$

$= A_1^2\Sigma_{1,1} + 2A_1A_2\Sigma_{1,2} + A_2^2\Sigma_{2,2}$

$= [A_1\ A_2]\begin{bmatrix}\Sigma_{1,1} & \Sigma_{1,2} \\ \underbrace{\Sigma_{1,2}}_{=\Sigma_{2,1}} & \Sigma_{2,2}\end{bmatrix}\begin{bmatrix}A_1 \\ A_2\end{bmatrix} = A\Sigma A^T$

So $Ax+b \sim N_1(A\mu+b, A\Sigma A^T) = N(A\mu+b, A\Sigma A^T)$

when $p=2$ $\quad (Ax+b)_1 = A_{1,1}x_1 + A_{1,2}x_2 + b$

$(Ax+b)_2 = A_{2,1}x_1 + A_{2,2}x_2 + b$

for transformed mean: (μ_1', μ_2'), same checking as above when $p=1$ (by linearity of $E[\cdot]$)

for transformed $\Sigma' = \begin{bmatrix}\Sigma_{1,1}' & \Sigma_{1,2}' \\ \Sigma_{2,1}' & \Sigma_{2,2}'\end{bmatrix}$

for $\Sigma_{1,1}' = Var((Ax+b)_1)$ and $\Sigma_{2,2}' = Var((Ax+b)_2)$,

same checking as above when $p=1$.

for $\Sigma_{1,2}' = Cov((Ax+b)_1, (Ax+b)_2) = Cov((Ax+b)_2, (Ax+b)_2) = \Sigma_{2,2}'$,

check: $Var((Ax+b)_1 + (Ax+b)_2)$

$= Var((Ax+b)_1) + Var((Ax+b)_2) + 2Cov((Ax+b)_1, (Ax+b)_2)$

$\Rightarrow Cov((Ax+b)_1, (Ax+b)_2) = \frac{1}{2}\big(Var((Ax+b)_1 + (Ax+b)_2)$
$\quad - Var((Ax+b)_1)$
$\quad - Var((Ax+b)_2)\big)$

$Var((Ax+b)_1 + (Ax+b)_2)$

$= Var((A_{1,1}+A_{2,1})X_1 + (A_{1,2}+A_{2,2})X_2 + 2b)$

$= (A_{1,1}+A_{2,1})^2 \Sigma_{1,1} + (A_{1,2}+A_{2,2})^2 \Sigma_{2,2} + 2(A_{1,1}+A_{2,1})(A_{1,2}+A_{2,2})\Sigma_{1,2}$

$Var((Ax+b)_1) = \Sigma'_{1,1} = A_{1,1}^2 \Sigma_{1,1} + 2A_{1,1}A_{1,2}\Sigma_{1,2} + A_{1,2}^2 \Sigma_{2,2}$

$Var((Ax+b)_2) = \Sigma'_{2,2} = A_{2,1}^2 \Sigma_{1,1} + 2A_{2,1}A_{2,2}\Sigma_{1,2} + A_{2,2}^2 \Sigma_{2,2}$

$\Sigma'_{1,2} = \Sigma'_{2,1}$

$$Cov((Ax+b)_1, (Ax+b)_2) = \frac{1}{2}\Big(2A_{1,1}A_{2,1}\Sigma_{1,1} + 2A_{1,2}A_{2,2}\Sigma_{2,2} + (\cancel{0A_{1,1}A_{1,2}} + 2A_{1,1}A_{2,2} + 2A_{2,1}A_{1,2} + \cancel{0A_{2,1}A_{2,2}})\Sigma_{1,2}\Big)$$

$$= A_{1,1}A_{2,1}\Sigma_{1,1} ① + A_{1,2}A_{2,2}\Sigma_{2,2} ④ + A_{1,1}A_{2,2}\Sigma_{1,2} ② + A_{2,1}A_{1,2}\Sigma_{1,2} ③$$

verify with

$$\underbrace{\begin{bmatrix} ? & \Sigma'_{1,2} \\ \Sigma'_{2,1} & ? \end{bmatrix}}_{=\Sigma'_{1,2}} = \begin{bmatrix} A_{1,1} & A_{1,2} \\ A_{2,1} & A_{2,2} \end{bmatrix} \begin{bmatrix} \Sigma_{1,1} & \Sigma_{1,2} \\ \Sigma_{2,1} & \Sigma_{2,2} \end{bmatrix} \begin{bmatrix} A_{1,1} & A_{2,1} \\ A_{1,2} & A_{2,2} \end{bmatrix}$$

$$= \begin{bmatrix} A_{1,1} & A_{1,2} \\ A_{2,1} & A_{2,2} \end{bmatrix} \begin{bmatrix} \Sigma_{1,1}A_{1,1} + \Sigma_{1,2}A_{1,2} & \Sigma_{1,1}A_{2,1} + \Sigma_{1,2}A_{2,2} \\ \Sigma_{2,1}A_{1,1} + \Sigma_{2,2}A_{1,2} & \Sigma_{2,1}A_{2,1} + \Sigma_{2,2}A_{2,2} \end{bmatrix}$$

$$\Sigma'_{1,2} = A_{1,1}A_{2,1}\Sigma_{1,1} ① + A_{1,1}A_{2,2}\Sigma_{1,2} ② \\ A_{1,2}A_{2,1}\underbrace{\Sigma_{2,1}}_{=\Sigma_{1,2}} ③ + A_{1,2}A_{2,2}\Sigma_{2,2} ④$$

Similarly for $\Sigma'_{2,1}$.

So proved for $p=2$ as well.

2.

2. $(*)$ **Conditional expectation for continuous and discrete random variables** [4 Points]

Let $(\Omega, \mathscr{F}, \mathbf{P})$ a probability space, and $X, Y : (\Omega, \mathscr{F}) \to (\mathbb{R}, \mathscr{B}(\mathbb{R}))$ random variables.

(a) Let X, Y be jointly continuous with joint density

$$f_{X,Y}(x,y) = e^{-y}\mathbb{1}_{\{0 \le x \le y\}}.$$

Calculate $\mathbf{E}[X|Y]$ and $\mathbf{E}[Y|X]$.

2.a. $f_X(x) = \int_x^\infty e^{-y}\,dy = [-e^{-y}]_x^\infty = e^{-x} \quad (x \geq 0)$

$f_Y(y) = \int_0^y e^{-y}\,dx = ye^{-y} \quad (y \geq 0)$

$$E[X|Y=y] = \frac{\int_{-\infty}^{\infty} xe^{-y}\mathbb{1}_{\{0\leq x\leq y\}}\,dx}{ye^{-y}} \quad (\text{assume } y \geq 0)$$

$$= \frac{\int_0^y xe^{-y}\,dx}{ye^{-y}} = \frac{e^{-y}[\frac{1}{2}x^2]_0^y}{ye^{-y}} = \frac{\left(\frac{e^{-y}y^2}{2}\right)}{ye^{-y}} = \frac{y}{2}$$

So $E[X|Y] = \frac{Y}{2} \quad (Y \geq 0)$

$$E[Y|X=x] = \frac{\int_{-\infty}^{\infty} ye^{-y}\mathbb{1}_{\{0\leq x\leq y\}}\,dy}{e^{-x}} \quad (\text{assume } x \geq 0)$$

$$\int_{-\infty}^{\infty} ye^{-y}\mathbb{1}_{\{0\leq x\leq y\}}\,dy \qquad = \frac{(x+1)e^{-x}}{e^{-x}} = x+1 \qquad \text{So } E[Y|X] = X+1 \quad (X \geq 0)$$

$$\int_x^\infty ye^{-y}\,dy = [-ye^{-y}]_x^\infty + \int_x^\infty e^{-y}\,dy$$

$$= xe^{-x} - [e^{-y}]_x^\infty$$

$$= (x+1)e^{-x}$$

(b) Let $r, p \in \mathbb{R}$ and $0 \le r \le p \le 1$ and $1 - 2p + r \ge 0$. The distributions of X and Y are given by

$$\mathbf{P}[X = 1, Y = 1] = r, \qquad \mathbf{P}[X = 0, Y = 1] = p - r,$$
$$\mathbf{P}[X = 1, Y = 0] = p - r, \qquad \mathbf{P}[X = 0, Y = 0] = 1 - 2p + r.$$

Determine the law of Y and the conditional expectation $\mathbf{E}[X|Y]$.

2.b. law of Y = pmf of Y

$P[Y=0] = (1-2p+r) + (p-r) = 1-p$

$P[Y=1] = (p-r) + r = p$

$$\text{pmf of } Y = \begin{cases} 1-p & y=0 \\ p & y=1 \\ 0 & \text{otherwise} \end{cases}$$

$$E[X|Y=0] = \frac{0(1-2p+r) + 1(p-r)}{1-p} = \frac{p-r}{1-p}$$

$$E[X|Y=1] = \frac{0(p-r) + 1(r)}{p} = \frac{r}{p}$$

$$\text{So } E[X|Y] = \begin{cases} \frac{p-r}{1-p} & Y=0 \\ \frac{r}{p} & Y=1 \end{cases}$$

(undefined otherwise)